Sections: B,C & D

Linear Algebra (MT-1004)

Homework#27

Instructor: Muhammad Ali

Multivariate Linear Regression

Question 1

A retail store wants to predict monthly sales (in thousands) based on two factors: the number of employees and the number of products displayed. The dataset is as follows:

Number of Employees	Products Displayed	Sales (Y in thousands)
10	15	30
3	7	7
2	10	14
5	8	22
10	12	16

- Write the design matrix for given data
- Compute the coefficients b (Bias) and w_1 (Weights)
- Perform a multivariate linear regression model on the provided dataset
- Predict the sales when there are 16 employees and 6 products displayed